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Research summary, October 2026

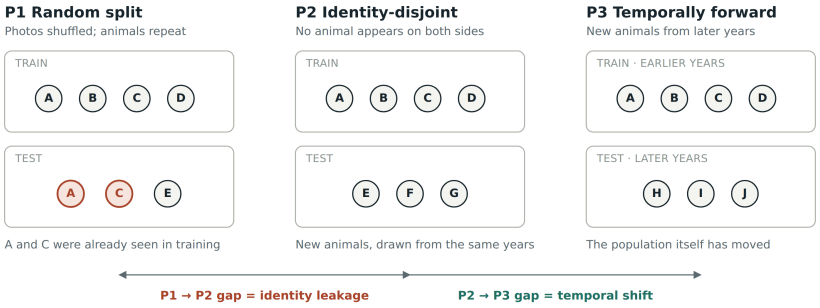
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The question

I build benchmarks, protocols and audits that test whether a model's reported performance survives deployment. My anchor is computer vision, and the same question carries into personalization, because both share one failure: when the same animals or the same people appear on both sides of a split, the reported number measures memory as much as generalisation. For me it began in 2019, with a model that estimates a cow's weight and breed from photographs, second runner-up for AI in Agriculture at Bangladesh's BASIS National ICT Awards. The award never asked whether it would still work a year later, on animals it had never seen, so I spent six years finding out.

BoviShift: separating what a benchmark measures from what it remembers

First author; manuscript available on request. With N. H. Tonmoy (Indiana University) and Dr. M. S. U. Miah (AIUB). I collected images of 2,657 bulls sold through one livestock marketplace between 2021 and 2026, with every image linked to the animal it shows. A random split lets the same animal appear in training and test; BoviShift adds an identity-disjoint protocol and a temporally forward one, so that an accuracy drop decomposes into identity leakage and temporal shift instead of being reported as one number. The 2024 cohort is held out of the primary temporal arm because its missingness is non-random. The decomposition carries to any data that re-observes the same individuals, from wildlife re-identification to recommender logs where the same users return. Two follow-up studies with the same team are under way.



Auditing LLM-based recommenders

Sole author, in preparation.
An LLM asked why it recommended an item will give a reason whether or not that reason drove the ranking. I am building an audit that measures explanation faithfulness, hallucinated item attributes, and whether either failure falls unevenly across user groups. It runs without GPU compute, so a team can repeat it on its own system.

Where the questions come from

Nearly eight years of production ML, now leading Graaho's ML and backend team. On Easymator, which measures construction drawings, we left wall detection manual, because a misdetected edge yields a confident wrong number nobody can see. On AlgoRec, a metered recommendation platform, no model went live until it beat a popularity baseline on AUC. On Koronik, an agent platform serving 5,000+ daily users at Islami Bank Bangladesh, the agent refuses and hands off to a person below a confidence threshold. Each rule is a small evaluation instrument built into a product. My research asks what it takes to make them general.

What I want to work on next

- Longitudinal, identity-aware benchmarks for vision, beginning with two BoviShift follow-up studies.
- The same decomposition for recommender benchmarks, where the same users return.
- Faithfulness and fairness audits for LLM recommenders that are cheap enough to become routine.
- Evaluation for agents that act on a user's behalf, where an unverified decision has a cost.

Earlier publications

M. B. Shagor, M. Z. H. Alvi, K. T. Hasan. CID: Cow Images Dataset for Regression and Classification. *ICCA '22*, doi:10.1145/3542954.3543018. **M. B. Shagor** et al. CAM – Categorized Affect Map. *ICCA 2020*, doi:10.1145/3377049.3377074. Full CV at mobasshirbhuiya.com.